CHAPTER 2

Random Variables

2–1 Concept of a Random Variable

The previous chapter deals exclusively with situations in which the number of possible outcomes associated with any experiment is finite. Although it is never stated that the outcomes had to be finite in number (because, in fact, they do *not*), such an assumption is implied and is certainly true for such illustrative experiments as tossing coins, throwing dice, and selecting resistors from bins. There are many other experiments, however, in which the number of possible outcomes is not finite, and it is the purpose of this chapter to introduce ways of describing such experiments in accordance with the concepts of probability already established.

A good way to introduce this type of situation is to consider again the experiment of selecting a resistor from a bin. When mention is made, in the previous chapter, of selecting a 1-Ω resistor, or a 10-Ω resistor, or any other value, the implied meaning is that the selected resistor is labeled "1 Ω" or "10 Ω." The actual value of resistance is expected to be close to the labeled value, but might differ from it by some unknown (but measurable) amount. The deviations from the labeled value are due to manufacturing variations and can assume any value within some specified range. since the actual value of resistance is unknown in advance, it is a *random variable.*

To carry this illustration further, consider a bin of resistors that are all marked "100 Ω." Because of manufacturing tolerances, each of the resistors in the bin will have a slightly different resistance value. Furthermore, there are an infinite number of possible resistance values, so that the experiment of selecting one resistor has an infinite number of possible outcomes. Even if it is known that all of the resistance values lie between 9.99 Ω and 100.01 Ω, there are an infinite number of such values in this range. Thus, if one defines a particular event as the selection of a resistor with a resistance of exactly 100.00 Ω, the probability of this event is actually zero. On the other hand, if one were to define an event as the selection of a resistor having a resistance

between 99.9999 Ω and 100.0001 Ω, the probability of this event is nonzero. The actual value of resistance, however, is a random variable that can assume any value in a specified range of values.

It is also possible to associate random variables with time functions, and, in fact, most of the applications that are considered in this text are of this type. Although Chapter 3 will deal exclusively with such random variables and random time functions, it is worth digressing momentarily, at this point, to note the relationship between the two as it provides an important physical motivation for the present study.

A typical random time function, shown in Figure 2–1, is designated as $x(t)$. In a given physical situation, this particular time function is only one of an infinite number of time functions that might have occurred. The collection of all possible time functions that might have been observed belongs to a random process, which will be designated as $\{x(t)\}$. When the probability functions are also specified, this collection is referred to as an *ensemble*. Any particular member of the ensemble, say $x(t)$, is a *sample function*, and the value of the sample function at some particular time, say t_1, is a random variable, which we call $X(t_1)$ or simply X_1. Thus, $X_1 = x(t_1)$ when $x(t)$ is the particular sample function observed.

A random variable associated with a random process is a considerably more involved concept than the random variable associated with the resistor above. In the first place, there is a different random variable for each instant of time, although there usually is some relation between two random variables corresponding to two different time instants. In the second place, the randomness we are concerned with is the randomness that exists from sample function to sample function throughout the complete ensemble. There may also be randomness from time instant to time instant, but this is not an essential ingredient of a random process. Therefore, the probability description of the random variables being considered here is also the probability description of the random process. However, our initial discussion will concentrate on the random variables and will be extended later to the random process.

From an engineering viewpoint, a random variable is simply a numerical description of the outcome of a random experiment. Recall that the sample space $S = \{\alpha\}$ is the set of all possible outcomes of the experiment. When the outcome is α, the random variable X has a value that we might denote as $X(\alpha)$. From this viewpoint, a random variable is simply a real-valued function defined over the sample space—and in fact the fundamental definition of a random variable is simply as such a function (with a few restrictions needed for mathematical consistency). For engineering applications, however, it is usually not necessary to consider explicitly the

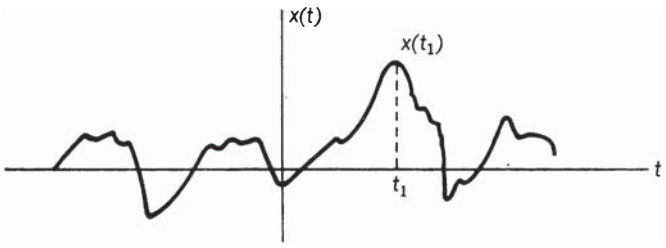


Figure 2–1 A random time function.

underlying sample space. It is generally only necessary to be able to assign probabilities to various *events* associated with the random variables of interest, and these probabilities can often be inferred directly from the physical situation. What events are required for a complete description of the random variable, and how the appropriate probabilities can be inferred, form the subject matter for the rest of this chapter.

If a random variable can assume any value within a specified range (possibly infinite), then it will be designated as a continuous random variable. In the following discussion all random variables will be assumed to be continuous unless stated otherwise. It will be shown, however, that discrete random variables (that is, those assuming one of a countable set of values) can also be treated by exactly the same methods.

2–2 Distribution Functions

To consider continuous random variables within the framework of probability concepts discussed in the last chapter, it is necessary to define the *events* to be associated with the probability space. There are many ways in which events might be defined, but the method to be described below is almost universally accepted.

Let X be a random variable as defined above and x be any allowed value of this random variable. The *probability distribution function* is defined to be the probability of the event that the observed random variable X is less than or equal to the allowed value x. That is,[1]

$$F_x(x) = \Pr(X \leq x)$$

Since the probability distribution function is a probability, it must satisfy the basic axioms and must have the same propenies as the probabilities discussed in Chapter 1. However, it is also a function of x, the possible values of the random variable X, and as such must generally be defined for all values of x. Thus, the requirement that it be a probability imposes certain constraints upon the functional nature of $F_x(x)$. These may be summarized as follows:

1. $0 \leq F_x(x) \leq 1 \quad -\infty < x < \infty$
2. $F_x(-\infty) = 0 \cdot F_x(\infty) = 1$
3. $F_x(x)$ is nondecreasing as x increases.
4. $\Pr(x_1 < X \leq x_2) = F_x(x_2) - F_x(x_1)$

Some possible distribution functions are shown in Figure 2–2. The sketch in (a) indicates a continuous random variable having possible values ranging from $-\infty$ to ∞ while (b) shows a continuous random variable for which the possible values lie between a and b. The sketch in (c) shows the probability distribution function for a discrete random variable that can assume only four possible values (that is, 0, a, b, or c). In distribution functions of this type it is important to

[1] The subscript X denotes the random variable while the argument x could equally well be any other symbol. In much of the subsequent discussion it is convenient to suppress the subscript X when no confusion will result. Thus $F_x(x)$ will often be written $F(x)$.

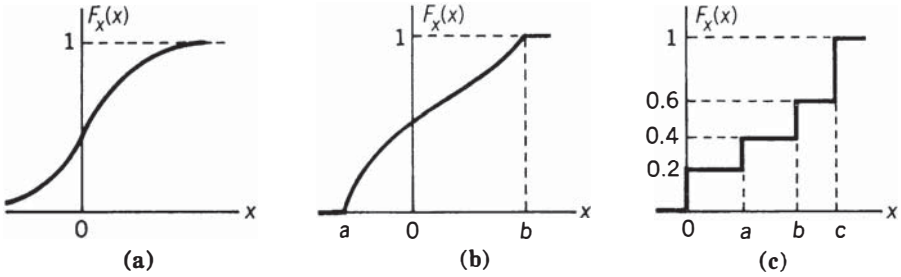


Figure 2–2 Some possible probability distribution functions.

remember that the definition for $F_x(x)$ includes the condition $X = x$ as well as $X < x$. Thus, in Figure 2–2(c), it follows (for example) that $F_x(a) = 0.4$ and not 0.2.

The probability distribution function can also be used to express the probability of the event that the observed random variable X is greater than (but not equal to) x. Since this event is simply the complement of the event having probability $F_x(x)$ it follows that

$$\Pr(X > x) = 1 - F_x(x)$$

As a specific illustration, consider the probability distribution function shown in Figure 2–3. Note that this function satisfies all of the requirements listed above. It is easy to see from the figure that the following statements (among many other possible statements) are true:

$$\Pr(X \leq -5) = 0.25$$

$$\Pr(X > -5) = 1 - 0.25 = 0.75$$

$$\Pr(X > 8) = 1 - 0.9 = 0.1$$

$$\Pr(-5 < X \leq 8) = 0.9 - 0.25 = 0.65$$

$$\Pr(X > 0) = 1 - \Pr(X \leq 0) = 1 - 0.5 = 0.5$$

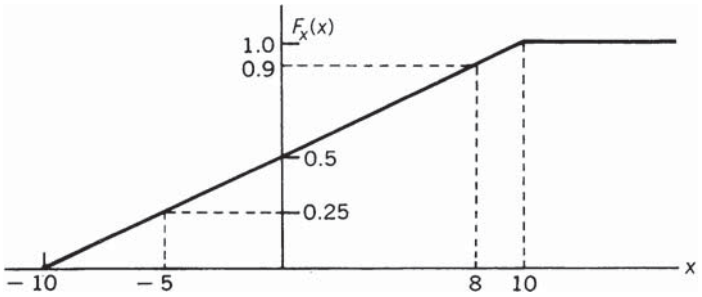


Figure 2–3 A specific probability distribution function.

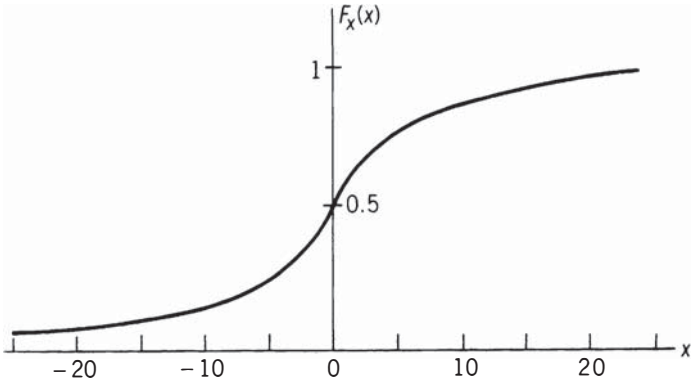


Figure 2–4 A probability distribution function with infinite range.

In the example above, all of the variation of the probability distribution function takes place between finite limits. This is not always the case, however. Consider, for example, a probability distribution function defined by

$$F_x(x) = \frac{1}{2}\left(1 + \frac{2}{\pi}\tan^{-1}\frac{x}{5}\right) \quad -\infty < x < \infty \tag{2–1}$$

and shown in Figure 2–4. Again, there are many different statements that can be made concerning the probability that the random variable X lies in certain regions. For example, it is straightforward to verify that all of the following are true:

$$\Pr(X \leq -5) = 0.25$$

$$\Pr(X > -5) = 1 - 0.25 = 0.75$$

$$\Pr(X > 8) = 1 - 0.8222 = 0.1778$$

$$\Pr(-5 < X \leq 8) = 0.8222 - 0.25 = 0.5722$$

$$\Pr(X > 0) = 1 - \Pr(X \leq 0) = 0.5$$

Exercise 2–2.1

A random experiment consists of flipping four coins and taking the random variable to be the number of heads.

a) Sketch the distribution function for this random variable.

b) What is the probability that the random variable is less than 3.5?

c) What is the probability that the random variable is greater than 2.5?

d) What is the probability that the random variable is greater than 0.5 and less than or equal to 3.0?

Answers: 15/16, 7/8, 5/16.

Exercise 2–2.2

A particular random variable has a probability distribution function given by

$$F_x(x) = 0 \qquad -\infty < x \leq 0$$
$$= 1 - e^{-2x} \qquad 0 \leq x < \infty$$

Find

a) the probability that $X > 0.5$

b) the probability that $X \leq 0.25$

c) the probability that $0.3 < X \leq 0.7$.

Answers: 0.3022, 0.3935, 0.3679

2–3 Density Functions

Although the distribution function is a complete description of the probability model for a single random variable, it is not the most convenient form for many calculations of interest. For these, it may be preferable to use the derivative of $F(x)$ rather than $F(x)$ itself. This derivative is called the *probability density function* and, when it exists, it is defined by[2]

$$f_x(x) = \lim_{e \to 0} \frac{F_x(x+e) - F_x(x)}{e} = \frac{dF_x(x)}{dx}$$

The physical significance of the probability density function is best described in terms of the probability element, $f_x(x)\,dx$. This may be interpreted as

$$f_x(x)\,dx = \Pr(x < X \leq x + dx) \tag{2–2}$$

[2] Again, the subscript denotes the random variable and when no confusion results, it may be omitted. Thus, $f_x(x)$ will often be written as $f(x)$.

Equation (2–2) simply states that the probability element, $f_x(x)\,dx$, is the probability of the event that the random variable X lies in the range of possible values between x and $x + dx$.

Since $f_x(x)$ is a density function and not a probability, it is not necessary that its value be less than l; it may have any nonnegative value.[3] Its general properties may be summarized as follows:

1. $f_x(x) \geq 0 \quad -\infty < x < \infty$
2. $\displaystyle\int_{-\infty}^{\infty} f_x(x)\,dx = 1$
3. $\displaystyle F_x(x) = \int_{-\infty}^{x} f_x(u)\,du$
4. $\displaystyle\int_{x_1}^{x_2} f_x(x)\,dx = \Pr(x_1 < X \leq x_2)$

As examples of probability density functions, those corresponding to the distribution functions of Figure 2–2 are shown in Figure 2–5. Note particularly that the density function for a discrete random variable consists of a set of delta functions, each having an area equal to the magnitude of the corresponding discontinuity in the distribution function. It is also possible to have density functions that contain both a continuous part and one or more delta functions.

There are many different mathematical forms that might be probability density functions, but only a very few of these arise to any significant extent in the analysis of engineering systems. Some of these are considered in subsequent sections and a table containing numerous density functions is given in Appendix B.

Before considering the more important probability density functions, however, let us look at the density functions that are associated with the probability distribution functions described in the previous section. It is clear from Figure 2–3 that the probability density function associated with this random variable must be zero for $x \leq -10$ and $x > 10$. Furthermore, in the interval

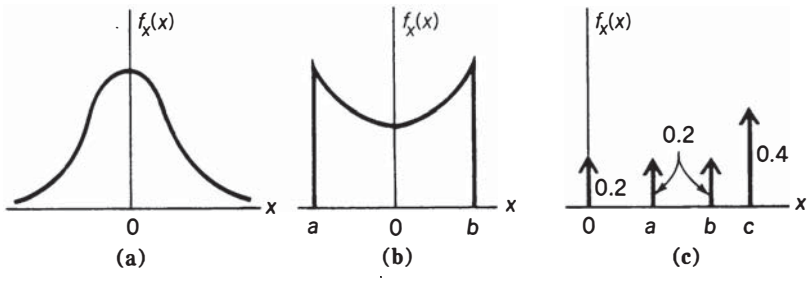


Figure 2–5 Probability density functions corresponding to the distribution functions of Figure 2–2.

[3] Because $F_x(x)$ is nondecreasing as x increases.

between -10 and 10 it must have a constant value since the slope of the distribution function is constant. Thus:

$$\begin{aligned} F_x(x) &= 0 && x \leq -10 \\ &= 0.05 && -10 < x \leq 10 \\ &= 0 && x > 10 \end{aligned}$$

This is sketched in Figure 2–6.

The probability density function corresponding to the distribution function of Figure 2–4 can be obtained by differentiating the distribution function of (2–1). Thus,

$$f_x(x) = \frac{dF_x(x)}{dx} = \frac{d}{dx}\left[\frac{1}{2} + \frac{1}{\pi}\tan^{-1}\frac{x}{5}\right] = \frac{5}{\pi}\left(\frac{1}{x^2 + 25}\right) \quad -\infty < x < \infty \tag{2–3}$$

This probability density function is displayed in Figure 2–7.

A situation that frequently occurs in the analysis of engineering systems is that in which one random variable is functionally related to another random variable whose probability density function is known and it is desired to determine the probability density function of the first random variable. For example, it may be desired to find the probability density function of a

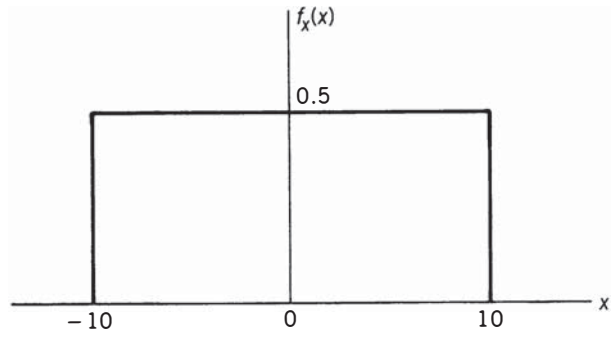


Figure 2–6 Probability density function corresponding to the distribution function of Figure 2–3.

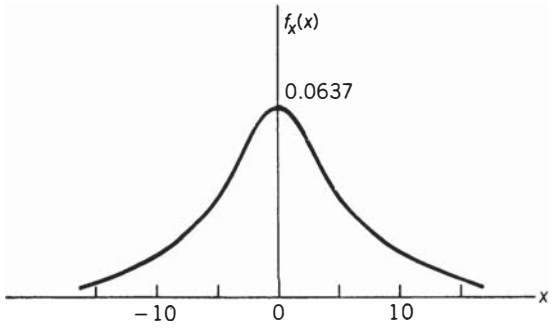


Figure 2–7 Probability density function corresponding to the distribution function of Figure 2–4.

power variable when the probability density function of the corresponding voltage or current variable is known. Or it may be desired to find the probability density function after some nonlinear operation is performed on a voltage or current. Although a complete discussion of this problem is not necessary here, a few elementary concepts can be presented and will be useful in subsequent discussions.

To formulate the mathematical framework, let the random variable Y be a single-valued, real function of another random variable X. Thus, $Y = g(X)$,[4] in which it is assumed that the probability density function of X is known and is denoted by $f_x(x)$, and it is desired to find the probability density function of Y, which is denoted by $f_y(y)$. If it is assumed for the moment that $g(X)$ is a monotonically increasing function of X, then the situation shown in Figure 2–8(a) applies. It is clear that whenever the random variable X lies between x and $x + dx$, the random variable Y will lie between y and $y + dy$. Since the probabilities of these events are $f_x(x)\,dx$ and $f_y(y)\,dy$, one can immediately write

$$f_Y(y)\,dy = f_X(x)\,dx$$

from which the desired probability density function becomes

$$f_Y(y) = f_X(x)\frac{dx}{dy} \tag{2–4}$$

Of course, in the right side of (2–4), x must be replaced by its corresponding function of y.

When $g(X)$ is a monotonically decreasing function of X, as shown in Figure 2–8(b), a similar result is obtained except that the derivative is negative. Since probability density functions must be positive, and also from the geometry of the figure, it is clear that what is needed in (2–4) is simply the absolute value of the derivative. Hence, for either situation

$$f_Y(y) = f_X(x)\left|\frac{dx}{dy}\right| \tag{2–5}$$

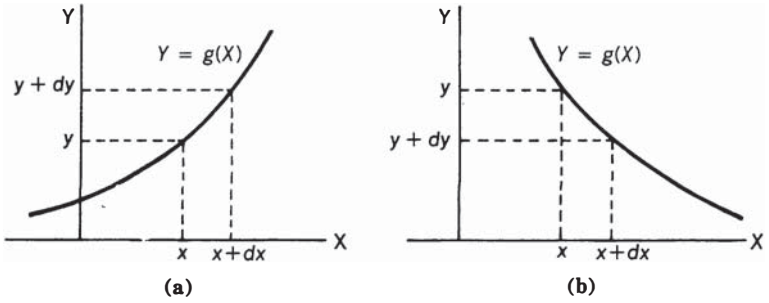


Figure 2–8 Transformation of variables.

[4] This also implies that the possible values of X and Y are related by $y = g(x)$.

To illustrate the transformation of variables, consider first the problem of scaling the amplitude of a random variable. Assume that we have a random variable X whose probability density function $fx(x)$ is known. We then consider another random variable Y that is linearly related to X by $Y = AX$. This situation arises, for example, when X is the input to an amplifier and Y is its output. Since the possible values of X and Y are related in the same way, it follows that

$$\frac{dy}{dx} = A$$

From (2–5) it is clear that the probability density function of Y is

$$f_Y(y) = \frac{1}{|A|} f_X\left(\frac{y}{A}\right)$$

Thus, it is very easy to find the probability density of any random variable that is simply a scaled version of another random variable whose density function is known.

Consider next a specific example of the transformation of random variables by assuming that the random variable X has a density function of the form

$$f_X(x) = e^{-x}u(x)$$

where $u(x)$ is the unit step starting at $x = 0$. Now consider another random variable Y that is related to X by

$$Y = X^3$$

Since y and x are related in the same way, it follows that

$$\frac{dy}{dx} = 3x^2$$

and

$$\frac{dx}{dy} = \frac{1}{3x^2} = \frac{1}{3y^{2/3}}$$

Thus, the probability density function of Y is

$$f_Y(y) = \frac{e^{-y^{1/3}}}{3} y^{-2/3}u(y)$$

There may also be situations in which, for a given Y, $g(X)$ has regions in which the derivative is positive and other regions in which it is negative. In such cases, the regions may be considered separately and the corresponding probability densities added. An example of this sort will serve to illustrate such a transformation.

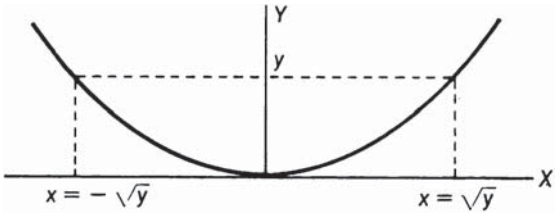


Figure 2–9 The square law transformation.

Let the functional relationship be

$$Y = X^2$$

This is shown in Figure 2–9 and represents, for example, the transformation (except for a scale factor) of a voltage random variable into a power random variable. Since the derivative, dx/dy, has an absolute value given by

$$\left|\frac{dx}{dy}\right| = \frac{1}{2\sqrt{y}}$$

and since there are two x-values for every y-value ($x = \pm\sqrt{y}$), the desired probability density function is simply

$$f_Y(y) = \frac{1}{2\sqrt{y}}\left[f_X(\sqrt{y}) + f_X(-\sqrt{y})\right] \quad y \geq 0 \tag{2–6}$$

Furthermore, since y can never be negative,

$$f_Y(y) = 0 \qquad y < 0$$

Some other applications of random variable transformations are considered later.

Exercise 2–3.1

The probability density function of a random variable has the form $f_X(x) = 5e^{-Kx}u(x)$, where $u(x)$ is the unit step function. Find

a) the value of K

b) the probability that $X > 1$

c) the probability that $X \leq 0.5$.

Answers: 0.0067, 5, 0.9179

Exercise 2–3.2

A random variable Y is related to the random variable X of Exercise 2–3.1 by

$$Y = 5X + 3$$

Find the probability density function of Y.

Answer: $e^{3-y}u(y-3)$

2–4 Mean Values and Moments

One of the most important and most fundamental concepts associated with statistical methods is that of finding average values of random variables or functions of random variables. The concept of finding average values for time functions by integrating over some time interval, and then dividing by the length of the interval, is a familiar one to electrical engineers, since operations of this sort are used to find the dc component, the root-mean-square value, or the average power of the time function. Such time averages may also be important for random functions of time, but, of course, have no meaning when considering a single random variable, which is defined as the value of the time function at a single instant of time. Instead, it is necessary to find the average value by integrating over the range of possible values that the random variable may assume. Such an operation is referred to as "ensemble averaging," and the result is the mean value.

Several different notations are in standard use for the mean value, but the most common ones in engineering literature are[5]

$$\overline{X} = E[X] = \int_{-\infty}^{\infty} xf(x)\,dx \tag{2–7}$$

The symbol $E[X]$ is usually read "the expected value of X" or "the mathematical expectation of X." It is shown later that in many cases of practical interest, the mean value of a random variable is equal to the time average of any sample function from the random process to which the random variable belongs. In such cases, finding the mean value of a random voltage or current is equivalent to finding its dc component; this interpretation will be employed here for illustration.

The expected value of any function of x can also be obtained by a similar calculation. Thus,

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)\,dx \tag{2–8}$$

[5]Note that the subscript X has been omitted from $f(x)$ since there is no doubt as to what the random variable is.

A function of particular importance is $g(x) = x^n$, since this leads to the general moments of the random variable. Thus,

$$\overline{X^n} = E[X^n] = \int_{-\infty}^{\infty} x^n f(x)\,dx \tag{2-9}$$

By far the most important moments of X are those given by $n = 1$, which is the mean value discussed above, and by $n = 2$, which leads to the mean-square value.

$$\overline{X}^2 = E[X^2] = \int_{-\infty}^{\infty} x^2 f(x)\,dx \tag{2-10}$$

The importance of the mean-square value lies in the fact that it may often be interpreted as being equal to the time average of the square of a random voltage or current. In such cases, the mean-square value is proportional to the average power (in a resistor) and its square root is equal to the rms or effective value of the random voltage or current.

It is also possible to define central moments, which are simply the moments of the difference between a random variable and its mean value. Thus the nth central moment is

$$\overline{(X - \overline{X})^n} = E\left[(X - \overline{X})^n\right] = \int_{-\infty}^{\infty} (x - \overline{X})^n f(x)\,dx \tag{2-11}$$

The central moment for $n = 1$ is, of course, zero, while the central moment for $n = 2$ is so important that it carries a special name, the *variance*, and is usually symbolized by σ^2. Thus,

$$\sigma^2 = \overline{(X - \overline{X})^2} = \int_{-\infty}^{\infty} (x - \overline{X})^2 f(x)\,dx \tag{2-12}$$

The variance can also be expressed in an alternative form by using the rules for the expectations of sums; that is,

$$E[X_1 + X_2 + \cdots + X_m] = E[X_1] + E[X_2] + \cdots + E[X_m]$$

Thus,

$$\begin{aligned} \sigma^2 &= E[(X - \overline{X})^2] = E[X^2 - 2X\overline{X} + (\overline{X})^2] \\ &= E[X^2] - 2E[X]\overline{X} + (\overline{X})^2 \\ &= \overline{X^2} - 2\overline{X}\,\overline{X} + (\overline{X})^2 = \overline{X^2} - (\overline{X})^2 \end{aligned} \tag{2-13}$$

and it is seen that the variance is the difference between the mean-square value and the square of the mean value. The square root of the variance, σ, is known as the *standard deviation.*

In electrical circuits, the variance can often be related to the average power (in a resistance) of the ac components of a voltage or current. The square root of the variance would be the value

indicated by an ac voltmeter or ammeter of the rms type that does not respond to direct current (because of capacitive coupling, for example).

To illustrate some of the above ideas concerning mean values and moments, consider a random variable having a uniform probability density function as shown in Figure 2–10. A voltage waveform that would lead to such a probability density function might be a sawtooth waveform that varied linearly between 20 and 40 V. The appropriate mathematical representation for this density function is

$$\begin{aligned} f(x) &= 0 \qquad -\infty < x \leq 20 \\ &= \frac{1}{20} \qquad 20 < x \leq 40 \\ &= 0 \qquad 40 < x < \infty \end{aligned}$$

The mean value of this random variable is obtained by using (2–7). Thus,

$$\overline{X} = \int_{20}^{40} x \left(\frac{1}{20}\right) dx = \frac{1}{20} \cdot \frac{x^2}{2}\bigg|_{20}^{40} = \frac{1}{40}(1600 - 400) = 30$$

This value is intuitively the average value of the sawtooth waveform just described. The mean-square value is obtained from (2–10) as

$$\overline{X}^2 = \int_{20}^{40} x^2 \left(\frac{1}{20}\right) dx = \frac{1}{20}\frac{x^3}{3}\bigg|_{20}^{40} = \frac{1}{60}(64 - 8)10^3 = 933.3$$

The variance of the random variable can be obtained from either (2–12) or (2–13). From the latter,

$$\sigma^2 = \overline{X}^2 - (\overline{X})^2 = 933.3 - (30)^2 = 33.3$$

On the basis of the assumptions that will be made concerning random processes, if the sawtooth voltage were measured with a dc voltmeter, the reading would be 30 V. If it were measured with an rms-reading ac voltmeter (which did not respond to dc), the reading would be $\sqrt{33.3}$ V.

As a second illustration of the determination of the moments of a random variable, consider the probability density function

$$f(x) = kx[u(x) - u(x - 1)]$$

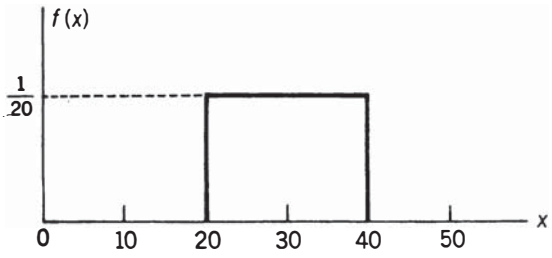


Figure 2–10 A uniform probability density function.

The value of k can be determined from the 0th moment of $f(x)$ since that is just the area of the density function and must be 1. Thus,

$$\int_0^1 kx\,dx = \frac{k}{2} = 1 \qquad \therefore k = 2$$

The mean and mean-square value of X may now be calculated readily as

$$\overline{X} = \int_0^1 x(2x)\,dx = 2/3$$

$$\overline{X}^2 = \int_0^1 x^2(2x)\,dx = 1/2$$

From these two quantities the variance becomes

$$\sigma^2 = \overline{X}^2 - (\overline{X})^2 = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{18}$$

Likewise, the 4th moment of X is

$$\overline{X}^4 = \int_0^1 x^4(2x)\,dx = \frac{1}{3}$$

and the 4th central moment is given by

$$\overline{(X - \overline{X})^4} = \int_0^1 \left(x - \frac{2}{3}\right)^4 (2x)\,dx = \frac{1}{135}$$

This latter integration is facilitated by observing that

$$\left(x - \frac{2}{3}\right)^4 x = \left(x - \frac{2}{3}\right)^5 + \frac{2}{3}\left(x - \frac{2}{3}\right)^4$$

Exercise 2–4.1

For the random variable of Exercise 2–3.1, find

a) the mean value of X

b) the mean-square value of X

c) the variance of X.

Answers: 2/25, 1/5, 1/5